

Static vs Contextual Embeddings for Lexical Ambiguity: A Word-in-Context Evaluation Method and Results Appendix

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Project repository: <https://github.com/Mr-Dark-debug/static-contextual-ambiguity>

AI-assisted wording: OpenAI Codex helped edit the explanatory prose in this appendix. A Claude rewrite supplied by the authors also informed Sections 2–10. We checked the wording against our saved results; the tables come from those results. Section 10 explains how we used each tool.

1 Research question and scope

This project asks whether a context-sensitive target representation distinguishes same-sense from different-sense uses more effectively than static alternatives. It compares type-level GloVe, a lexical GloVe context mean, and an unfine-tuned BERT-base target representation. It is a controlled representation probe, not a claim about fine-tuned task state of the art.

2 Dataset provenance and audit

We used the official SuperGLUE v2 release of WiC, available at <https://dl.fbaipublicfiles.com/glue/superglue/data/v2/WiC.zip>. The WiC authors distribute it under CC BY-NC 4.0.

Table 1: Validated release counts and span diagnostics.

Split	Rows	Positive	Negative	Unlabeled	Lemma mismatch	Repeated surface
Train	5,428	2,714	2,714	0	1,852	93
Validation	638	319	319	0	251	1
Test	1,400	0	0	1,400	521	20

We checked the row counts and target positions in every split. Each target span has a start and end position and contains at least one character. IDs are unique within each split. Training contains four symmetric duplicate pairs; validation contains none. We found no such pair shared between training and validation. The test labels are hidden, so we checked the split but did not score it.

3 Representations

3.1 Static baselines

Both static comparisons use the uncased GloVe release with 300 numbers per word. The target-only comparison looks up the WiC lemma in each sentence. Because the lemma is the same on both sides, its similarity is always 1 when a vector is available. GloVe has no vector for 18 training lemmas; we replaced those missing scores with the median of the available training scores.

For the nearby-word comparison, we averaged the GloVe vectors of words around the marked target, leaving out the target itself. We tried two, five, and ten words on either side, as well as all words in the sentence. If none of the surrounding words had a GloVe vector, we used the target lemma as a fallback. That happened once in training for each window size and never in validation.

3.2 Contextual target

We used the pretrained `google-bert/bert-base-uncased` checkpoint without fine-tuning it. The tokenizer tells us which pieces of text belong to the marked word. We averaged the BERT vectors for all pieces that overlap that word. One pass through BERT gave us each of its 12 layers and the average of the last four layers. We saved these representations for the comparisons below.

4 Selection and evaluation protocol

We made every model choice using training data. With random seed 2026, we split the 5,428 balanced training pairs into 4,342 pairs for tuning and 1,086 for an internal check, keeping the same class balance in both groups. We chose a similarity cutoff for each candidate on the tuning group using macro F1. Then we compared candidates on the internal group, using macro F1 first, followed by accuracy, ROC-AUC, and the simpler representation if needed. We fixed the winners before looking at validation results and refitted their cutoffs on all 5,428 training pairs.

Table 2: Training-only candidate selection.

Family	Candidate	Tune macro F1	Holdout macro F1	Selected
Static context	glove_context_2	0.5840	0.5961	yes
Static context	glove_context_5	0.5883	0.5928	no
Static context	glove_context_10	0.5917	0.5863	no
Static context	glove_context_sentence	0.5930	0.5845	no
BERT	layer_12	0.6789	0.7145	no
BERT	mean_last_four	0.6891	0.7247	yes

5 Primary validation results

We predict the same meaning when a pair’s similarity reaches its chosen cutoff. For precision and recall, same meaning is the positive class. The confidence intervals come from 1,000 resamples of the validation pairs, using a fixed seed.

Table 3: Primary results on 638 balanced validation pairs.

System	Threshold	Accuracy	Accuracy 95% CI	Macro F1	Precision	Recall	ROC-AUC
GloVe target	1.0000	0.5000	[0.4639, 0.5392]	0.3333	0.5000	1.0000	0.5000
GloVe context (± 2)	0.6506	0.5549	[0.5156, 0.5956]	0.5511	0.5670	0.4639	0.5800
BERT mean last four	0.5813	0.6708	[0.6364, 0.7085]	0.6673	0.6416	0.7743	0.7163

The selected BERT representation got 428 pairs right, while the nearby-word GloVe comparison got 354. Both were right on 250 pairs. BERT alone got 178 right, GloVe alone got 104 right, and both were wrong on 106.

Table 4: Paired primary-system differences with 95% intervals.

Contrast	Metric	Estimate	95% interval
Target GloVe minus nearby GloVe	Accuracy	-0.0549	[-0.1176, 0.0094]
Target GloVe minus nearby GloVe	Macro F1	-0.2178	[-0.2661, -0.1710]
Target GloVe minus nearby GloVe	ROC-AUC	-0.0800	[-0.1273, -0.0378]
Target GloVe minus BERT	Accuracy	-0.1708	[-0.2195, -0.1238]
Target GloVe minus BERT	Macro F1	-0.3340	[-0.3739, -0.2953]
Target GloVe minus BERT	ROC-AUC	-0.2163	[-0.2588, -0.1775]
Nearby GloVe minus BERT	Accuracy	-0.1160	[-0.1693, -0.0658]
Nearby GloVe minus BERT	Macro F1	-0.1161	[-0.1690, -0.0664]
Nearby GloVe minus BERT	ROC-AUC	-0.1363	[-0.1912, -0.0811]

The intervals for the difference between GloVe context and BERT exclude zero for accuracy, macro F1, and ROC-AUC. BERT therefore performed better on these validation pairs under our comparison procedure. We would need other datasets to know whether the same ranking holds elsewhere.

6 Secondary BERT layer analysis

We also looked at the BERT layers individually. We fitted each layer’s cutoff on training pairs. These validation results describe the layers; they did not change our chosen representation.

Table 5: Individual BERT target layers.

Layer	Threshold	Accuracy	Macro F1	ROC-AUC
1	0.8546	0.5705	0.5685	0.5900
2	0.8145	0.6019	0.6014	0.6187
3	0.7590	0.6113	0.6113	0.6357
4	0.7330	0.6317	0.6314	0.6575
5	0.6949	0.6458	0.6451	0.6768
6	0.6564	0.6442	0.6419	0.6853
7	0.6299	0.6489	0.6467	0.6946
8	0.6039	0.6599	0.6558	0.7036
9	0.5728	0.6661	0.6624	0.7093
10	0.5625	0.6599	0.6543	0.7197
11	0.5947	0.6536	0.6508	0.7096
12	0.4942	0.6364	0.6308	0.6971

Accuracy generally rises from layer 1 to layer 9, then falls through layer 12. Layer 10 has the highest ROC-AUC of the individual layers. This pattern fits with the last-four-layer average chosen during training, but we did not use validation to choose it.

7 Error analysis

The groups above do not overlap. On 54 of the 106 pairs missed by both main systems, the target-only GloVe diagnostic was also wrong; on the other 52 it happened to be right. BERT missed 210 pairs in total. Thirty-three of those scores were within 0.025 of its cutoff, while 53 were in the quarter of errors furthest from the cutoff.

BERT recovered 77.43% of same-meaning pairs and 56.74% of different-meaning pairs. Nearby-word GloVe recovered 46.39% and 64.58%, respectively. In saved examples, BERT placed a same-meaning anatomical *ball/balls* pair far apart, but placed different meanings of *engrave* and *map* close together. It also linked the same meaning in *rust remover* and *paint remover*, despite their different wording. Fine differences in meaning may explain some errors, but these examples alone cannot establish the cause.

8 Reproducibility record

We ran BERT inference on an NVIDIA GeForce RTX 3050 Ti Laptop GPU. The repository contains the locked software versions, configuration, scripts, and saved results needed to repeat the comparison.

9 Limitations

We tested one English dataset, so the result may change on other tasks or languages. Our intervals reflect variation among these validation pairs, not uncertainty from the dataset design or our choice of models. We used an older frozen BERT base checkpoint. Our simple GloVe context average does not model word order or syntax. A single similarity cutoff may also miss useful detail. We did not score the test split because its labels are unavailable.

10 Author contributions and use of AI tools

We chose WiC, planned the three comparisons, loaded the pretrained GloVe and BERT models, and selected the settings and cutoffs from training data. We ran the dataset checks, model inference, and validation analysis, then checked the tables and figures against the saved results. We did not train or fine-tune model weights.

We chose the examples, explanations, and graphs for the poster and assembled it in LaTeX. We used Canva to explore the visual layout and GLM to help write some of the LaTeX layout code. OpenAI Codex helped with basic coding and debugging while we worked through the code together. It also helped draft and edit the appendix, including this disclosure, and checked the PDFs. A rewrite produced with Claude helped us make Sections 2–10 easier to read. We checked these suggestions against our experiment and take responsibility for the code, data, results, wording, and final submission. We will identify any further AI-generated passages in the appendix if we add them before submission.

References

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